BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 210 · Debugging and error correction

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Write −1 in 8-bit two's complement.

Q2. What is the last index in a four-item array?

Q3. What precondition does binary search need?

Q4. What happens if code reads a[3] from a three-element C# array?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

off-by-one error — Loop repeats once too many or few times.

exception — Runtime failure or signal raised by an invalid operation.

trace — Record changing execution state.

precondition — Input requirement an algorithm assumes.

termination — An algorithm eventually ends.

New learning

Check valid indexes — A three-element array has valid indexes 0, 1 and 2.

Follow the condition — When i is 3, i<=a.Length remains true and attempts to read a[3], causing an exception.

Fix and retest — Use i<a.Length, which stops before i becomes an invalid index.

Worked example

Correct a loop summing [2,4] that uses <= length.

1. Predict — The array's valid indexes are 0 and 1, not 2.

2. Explain — Use i < values.Length and the total is 6.

We Do · Practise with guidance

Binary search uses low=mid when a[mid]<target. Why may it loop forever?

```
int mid=(low+high)/2;
if(a[mid]<target) low=mid;
```

Predict

With low=3 and high=4, midpoint remains 3 and low=mid fails to shrink the interval.

Explain

Use low=mid+1 to exclude the checked midpoint and guarantee progress.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Fix the loop condition i<=a.Length when visiting every array item.

Question 2

When a[mid]>target, what should binary search do to high?

Question 3

Choose normal, boundary and erroneous indexes when a method accepts only indexes 0–4.

Exit Ticket · Answer before checking

Using low=mid+1 after a[mid]<target can prevent non-shrinking binary-search intervals.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 11111111.

Q2. 3.

Q3. Sorted input.

Q4. An IndexOutOfRangeException.

You Do pages · Independent questions

Q1. Change to i<a.Length.

Q2. Set high=mid-1 to remove the checked midpoint.

Q3. Normal index 2; boundary indexes 0 or 4; erroneous indexes −1 or 5, which must be rejected or handled.

Exit Ticket · Model answer

True. It excludes the checked midpoint.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: